FROM THE

EDITOR-IN-CHIEF

Stress, as a factor impacting health, has been around for long in our lexicon. But it was till recently seen only as a minor player among an array of more ominous physiological factors that caused major illnesses. That perception has changed, and quite radically, with new scientific studies showing that stress could be the mother of all ailments. It is now seen as a fundamental disorder of our mental and physical metabolism that has a debilitating and potentially lethal effect—as a causative factor for heart attacks, hypertension, cancer, brain stroke, diabetes, liver cirrhosis, obesity, osteoporosis, infertility, depression and even suicide. Indeed, the World Health Organization has responded to this by declaring stress to be the "health epidemic of the 21st century".

In the US and other advanced nations, stress has already been recognised as a major disorder. The American Institute of Stress recently estimated that about 75-90 per cent of all doctor visits are for stress-related problems. India, too, is now falling victim to this silent killer. A December 2020 study of 10,000 Indians conducted by The Centre of Healing in Delhi found that 74 per cent Indians suffered from stress, while 88 per cent reported anxiety—a related effect with many overlaps. The data also showed that nearly 70 per cent stress therapists documented an increase in the number of patient visits; 55 per cent said there had been an increase in first-time clients after the Covid-19 pandemic. "The pandemic has increased stress and we are finding that stress has a greater impact on health than previously seen," says Dr Ashok Seth, chairman, Fortis Escorts Heart Institute. "There is an uptick in sudden heart attacks, for example, with stress being a significant contributing factor." Another study of 41,600 professionals in 13 countries by Oracle and Workplace Intelligence in October 2021 found Indian professionals to be the most stressed at the workplace—with 91 per cent of the 1,100 respondents saying they were highly stressed out by work-life situations compared to a global average of 80 per cent.



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How did we go from seeing stress as an occasional outbreak of anxiety or even panic to thinking of it as a fundamental and structural factor in our lives, colouring everything, and contributing to all manner of ailments? It's quite simply the more microscopic understanding of bodily processes that modern biological sciences continue to grant to us. Senior Associate Editor Sonali Acharjee, who wrote this week's cover story, spoke to a range of researchers and therapists to understand the phenomenon to its fullest extent. It meant taking a deep dive into the science behind how our body handles stress.

The normal functioning of stress is, in fact, what allows us to survive and thrive. Every perception of a threat or an impending adverse event is meant to trigger what is called a fight-or-flight syndrome. It sets off a chain of neural-hormonal reactions that gives us a bionic burst of energy, heightened pulse, sweating, and the muscles involved in running getting into a hyperactive zone to enable us to meet the challenge. The alarm is only one end of its typical functional spectrum. Stress also allows us to respond positively to challenges by being alert, focused and thus productive—a virtuous territory known as 'eustress'.

Some obvious stressful situations are losing a loved one, disease, natural calamities, war or any conflict that poses a consistently clear and present danger to life. However, the problem is that contemporary life has added many more triggers to the usual ones. Some may seem mundane: the everyday stress of commuting on crowded trains or navigating peak-hour jams, including the increasing noise pollution. Then there is economic stress, including job insecurity, long working hours, layoffs, and paying off financial debt. Ever so often, these triggers hunt in packs and keep individuals on edge all the time, leading to what is known as chronic stress.

Researchers worldwide, including in India, have begun to understand why chronic stress is so harmful. The endocrine system controls the physical impact of stress, particularly the hypothalamus-pituitary-adrenal axis in our body. When the brain senses a threat, it stimulates the hypothalamus to secrete hormones acting like chemical messengers to activate the adrenal glands above our kidneys. The adrenal glands respond by releasing the stress hormone—cortisol—which then powers up our body's in-built flight-or-fight response. After the threat is over, its levels are meant to drop in the bloodstream.

In normal circumstances, this process is responsible for protecting our bodies against acute stressful situations. But under prolonged stress, the natural thermostat that turns off the secretion of cortisol malfunctions and our body remains heated up with chemicals to combat everyday stress. "When cortisol was first discovered, it was a breakthrough," says Delhi-based neurologist Dr P.N. Renjen. "But we soon realised steroids are a double-edged sword. If you have too much of them in your body, instead of protecting you, they start to do the opposite because they become inflammatory when given in excess. Too much cortisol is now seen as a precursor to several neurological diseases, among other ailments."

Fortunately, new research is also shedding fresh light on how to manage stress. No longer are medicines and therapy the only, or the first, answer. From crystal healing to meditation, *pranayama* to walking, building a positive social network to journal-keeping, gardening to doing jigsaw puzzles—doctors now advise patients on various innovative ways to reduce stress and relax. All these are designed to make individuals look for solutions rather than get locked into an automated stress response when confronted with challenges. Also in our cover story are some of these tips on managing stress and leading a healthy life. Read and Relax.

(Aroon Purie)